# Poisson Regression

## Intro + Risk Estimation

Mabel Carabali

EBOH, McGill University

(updated: 2026-11-04)

### Expected competencies

- Knowledge about linear, logistic, and binomial regressions.
- Knowledge about count data and Poisson distribution.

# Objectives

- Revise Poisson distribution and model assumptions
- Provide tools to estimate Risk Ratios, Risk Difference and Rate Ratios using Poisson regressions

# What have we covered so far?

**Type of regression and Interpretations**

| | Model | Interpretation of $b_1$ |
|---|---|---|
| Linear | $y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$ | Increase in outcome $y$ mean value (continuous variable) per unit increase in $x_1$, adjusted for all other variables in the model |
| Logistic | $\log(\text{odds}) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$ | Increase in the log odds of the outcome per unit increase in $x_1$, adjusted for all other variables in the model |
| Cox | $\log(\text{hazard}) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$ | Increase in the log hazard of the outcome per unit increase in $x_1$, adjusted for all other variables in the model |
| Poisson | $\log(\text{rate}) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$ | Increase in the log rate of the outcome per unit increase in $x_1$, adjusted for all other variables in the model |

SZKLO, M.; NIETO, F. J. **Epidemiology**. Burlington, Massachusetts: Jones & Bartlett Learning, 2019. V. Fourth edition.(Table 7-15)

SZKLO, M.; NIETO, F. J.Epidemiology. Burlington, Massachusetts: Jones & Bartlett Learning, 2019. V. Fourth edition. (Table 7-15)

# Poisson distribution



# Poisson distribution

The Poisson distribution gives the probability mass function (discrete) of all possible numbers of new cases, from 0 to $\infty$.

For a Poisson random variable, the **variance = mean = E(Y) = $\lambda$**

where $\lambda$ = expected number of cases in a given time period

If $Y_i$ = # of new cases with $\lambda$ and

$$Y_i \sim Poisson(\lambda)$$

then the probability mass function that $Y = k$ (a particular count) is:

$$P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

# Poisson distribution

A special case of the binomial:

- with trials $n \to \infty$ and $P$(success in any trial) $\to 0$
- If $p$ is small, binomial P(k successes) $\approx$ Poisson, P(k with $\lambda = np$)

**Example:**

Consider binomial distribution with probability = 0.01. What is the probability of 8 occurrences in 1000 tries?

**Binomial**

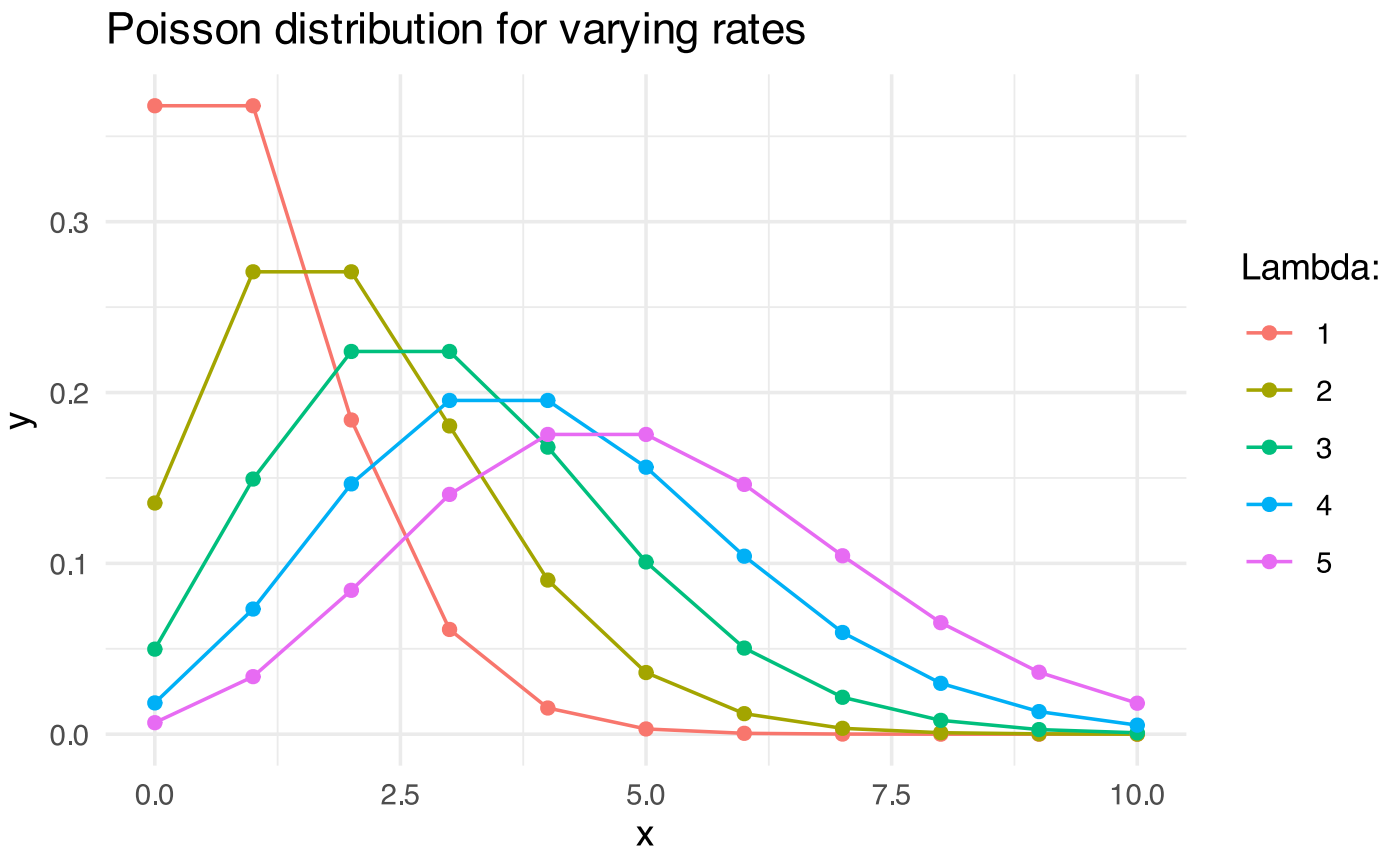
```
# binomial
dbinom(8,1000,0.01)
```

```
## [1] 0.1128241
```

**Poisson**

```
# poisson
dpois(8,1000*.01) # lambda = n*p
```

```
## [1] 0.112599
```




- As λ increases Poisson distribution approximates to a normal distribution.
- When $n$ is large and $p$ is very small, Poisson distribution approximates the binomial distribution.

# Poisson Generalized linear models

Framework for statistical analysis (linear & logistic regression special cases) that involve:

1. A vector of outcome data y = (y1,..., yn)
2. A matrix of predictors $X$ and vector of coefficients $\beta$, forming a linear predictor vector $X\beta$
3. A link function $g$, yielding a vector of transformed data $\hat{y} = g^{-1}(X\beta)$
4. Possibly other parameters, such as variances, overdispersions, and cutpoints, involved in the predictors, link function, and data distribution

Poisson regression model is **NOT** the same as taking the $log(Y)$ and then performing linear regression

- In a Poisson regression, the left-hand side is $E[log(Y|x)]$, not the log of the expectation. Poisson model $\neq$ log transformed linear model.

$$E[log(Y)|x] \neq log(E[Y|x])$$

# Poisson distribution

**What's the advantage of working with $log(E[Y|X])$ when dealing with counts?**

If the log of the expected count $log(E[Y|X])$ is modeled, no danger of negative predicted counts

- The antilog of any negative regression values correspond to expected counts between 0 and 1

```
exp(-.001)
```

```
## [1] 0.9990005
```

```
round(exp(-10)); #exp(-10);
```

```
## [1] 0
```

**... and this is also useful for predicting probabilities!**

# Poisson Generalized Linear Models

A GLM with a log link and a Poisson distribution:

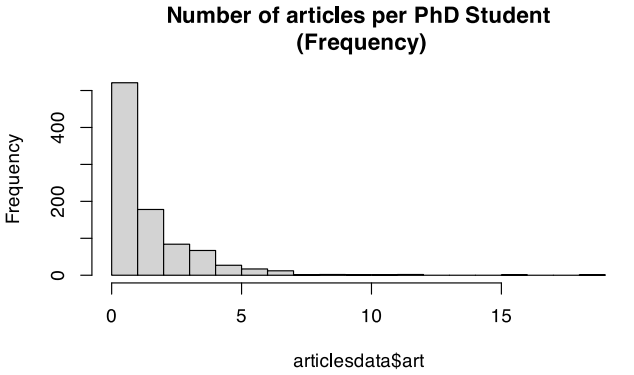
- Used to model count variables
- Assumes that the **conditional means equal the conditional variances**.
- A natural model for rates in data sets with person time (e.g: Mortality).
- Under predicts zero counts and over predicts the first set of counts
- Assumes *same rate of outcome among the individual observations* but incorporating "observed heterogeneity" relax this assumption.

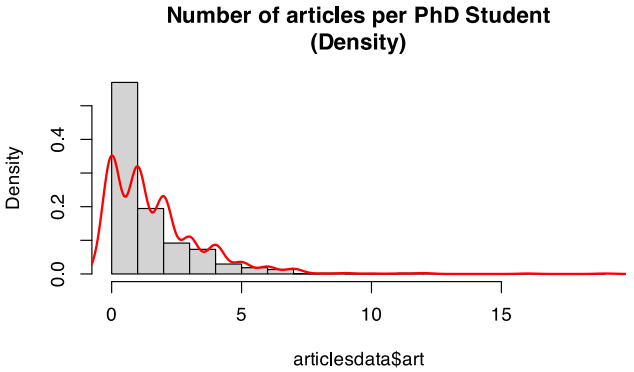
# Illustration:

**`bioChemists` (or Articles) dataset:** It tracks the publication productivity of 915 biochemistry doctoral students. Derived from a seminal study by J. Scott Long (1990) dataset found here.

| Characteristic | N = 915[1] |
|---|---|
| art | |
| Median (Q1, Q3) | 1 (0, 2) |
| Min, Max | 0, 19 |
| mar | |
| Married | 606 (66%) |
| Single | 309 (34%) |
| fem | |
| Men | 494 (54%) |
| Women | 421 (46%) |
| phdage | |
| Median (Q1, Q3) | 35.0 (32.0, 37.0) |
| Min, Max | 28.0, 42.0 |

[1] n (%)



# Poisson Model

```
pois0<- glm(art ~ 1, family = poisson(link="log"), data = articlesdata)
mean(articlesdata$art)
```

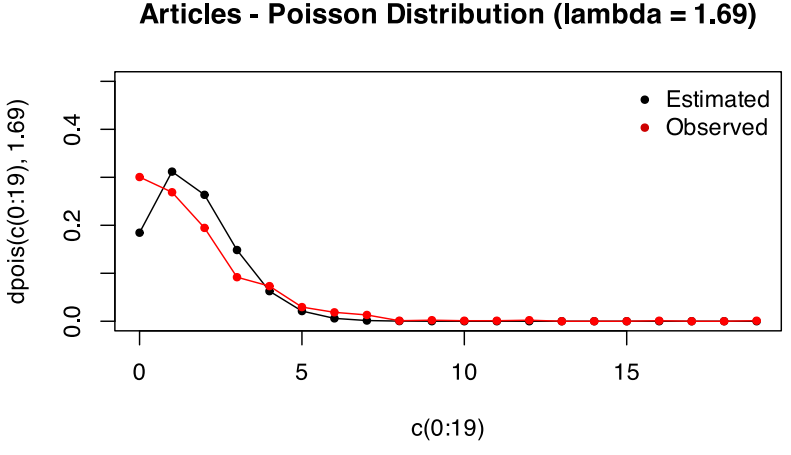
```
## [1] 1.692896
```

On the log scale (log(rate))

| | Est. | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 0.53 | 0.48 | 0.58 | 20.72 |
| Standard errors: MLE | | | | |

Multiplicative Scale (rate)

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 1.69 | 1.61 | 1.78 | 20.72 |
| Standard errors: MLE | | | | |




- Under predicts zero counts
- over predicts the first set of counts

# Poisson Model with Covariates

$\lambda =$ expected number of cases in a given time period/space. If the follow-up time is not specified at all, the assumption is that person-time is 1 for everyone, and `exp(β)` is a **RISK ratio** not the Rate Ratio.

```
pois1<- glm(art ~ fem + mar,
        family = poisson(link="log"),
        data = articlesdata)
jtools::summ(pois1, confint=T, exp=T, model
```

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 1.89 | 1.77 | 2.03 | 18.19 |
| **femWomen** | 0.79 | 0.71 | 0.87 | -4.49 |
| **marSingle** | 0.97 | 0.87 | 1.09 | -0.50 |

Standard errors: MLE

Note what happened to the intercept

```
pois2<- glm(art ~ fem + mar +
        offset(log(phdage)),
        family = poisson(link="log"), data
jtools::summ(pois2, confint=T, exp=T, model
```

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 0.05 | 0.05 | 0.06 | -82.70 |
| **femWomen** | 0.78 | 0.70 | 0.87 | -4.64 |
| **marSingle** | 0.98 | 0.88 | 1.10 | -0.29 |

Standard errors: MLE

# Poisson Model with Offset

Can't model only the number of outcomes as the exposure time varies. Need to model rates **(outcome/person-time)**.

So, instead of typical:

$$\log \mu_x = \beta_0 + \beta_1 x$$

(where $\mu_x$ is the expected count for those with covariate x), you need:

$$\log \frac{\mu_x}{t_x} = \beta_0' + \beta_1' x$$

(where $t_x$ is the exposure time for those with covariate x). Can be rewritten:

$$\log \mu_x = \log t_x + \beta_0' + \beta_1' x$$

where $\log t_x$ = the offset (denominator of the rate)

# Poisson Model with Offset



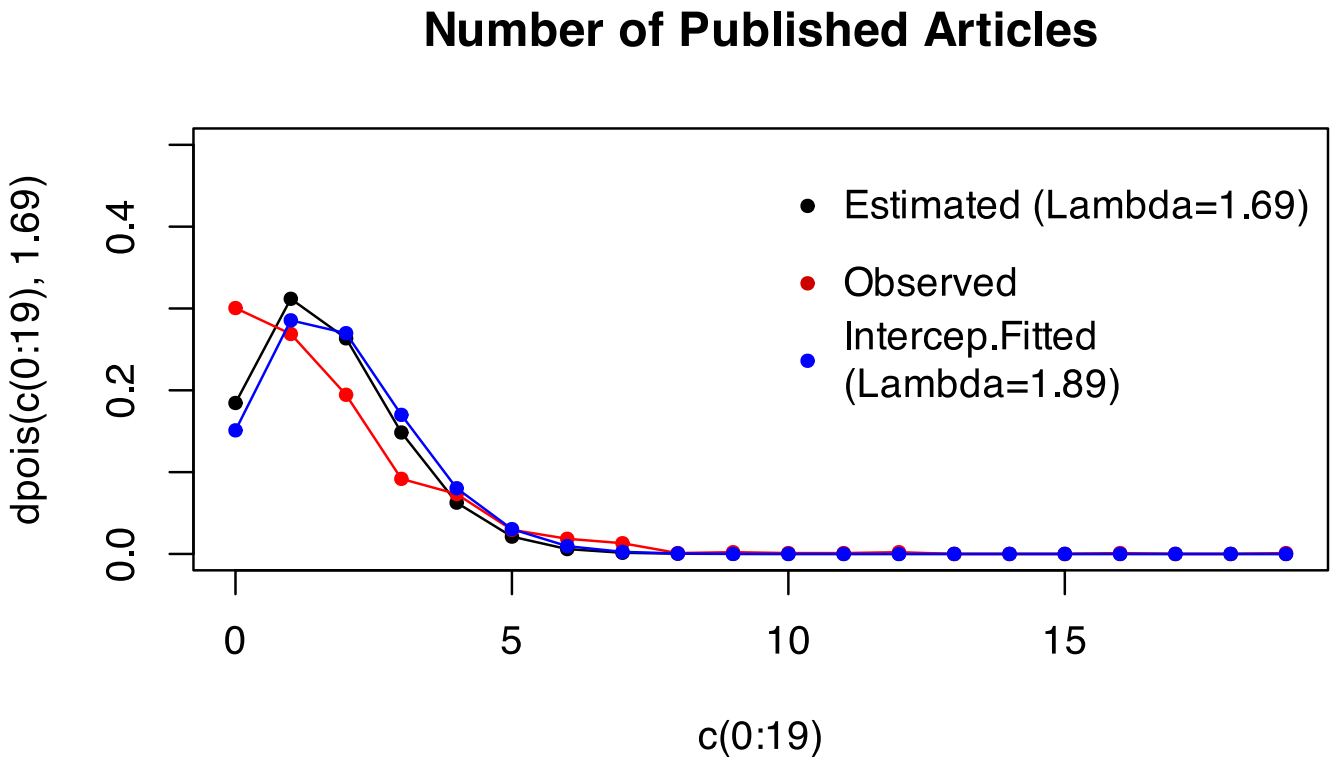
**OFFSET:** The parameter representing the size, exposure or measurement time, or population size of each observational unit.

- Including the offset allow varying size, exposure or measurement time, or population size for the observations.

- Models without the offset assume everyone has the same duration of exposure or that the denominator is same for everyone constrained to 1.

- Adding an offset allows estimation of rates allowing for varying exposure or person-time.

# Poisson Model with Covariates




- To get the RATE ratio instead of RISK ratio, we provide the follow-up time the offset
- Depending on the influence of the denominator, the rate and risk be similar or not.

# Poisson Model: Assumptions

- (1) That the outcome is a count with no negative values;
- (2) Independence of observations (the occurrence of one event does not affect the probability of another);
- (3) a linear relationship between predictors and outcome (log of the expected count), and
- (4) equi-dispersion or homoscedasticity, where **variance = mean = E(Y) =** $\lambda$;
- (5) constant rates

# Poisson and Negative Binomial models

- The usual link function $g$ used is the logarithmic, so that $g(u) = exp(u)$ transforms a continuous linear predictor $X_i\beta$ to a positive $\hat{y}$ (rate ratio)

- The simplest regression model for count data is

$$y_i \sim Poisson(e^{X_i\beta})$$

The linear predictor $X_i\beta$ is the logarithm of the expected value of measurement $y_i$, where $sd(y_i) = \sqrt{E(y_i)}$

- When $variance \neq mean$, we can use the **negative binomial distribution**: including an extra parameter to capture overdispersion, that is, variation in the data beyond what would be predicted from the Poisson distribution alone. More of this in 705

# How to "measure" the overdispersion?

### Basics:

```
mean(articlesdata$art); var(articlesdata$art)
```

```
## [1] 1.692896
```

```
## [1] 3.709742
```

Or using the `overdispersiontest`

```
library(AER)
# Run the overdispersion test
dispersiontest(pois2, alternative = "greater")
```

```
##
##     Overdispersion test
##
## data:  pois2
## z = 4.6335, p-value = 1.798e-06
## alternative hypothesis: true dispersion is greater than 1
## sample estimates:
## dispersion
##   2.192731
```

# Negative Binomial models

```
#Note the use of glm.nb +offset() (1)
pois2a<- glm.nb(art ~ fem + mar +
        offset(log(phdage)),
        data = articlesdata)
jtools::summ(pois2a, confint=T, exp=T, mode
```

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 0.06 | 0.05 | 0.06 | -57.08 |
| **femWomen** | 0.78 | 0.67 | 0.90 | -3.34 |
| **marSingle** | 0.98 | 0.84 | 1.14 | -0.28 |

Standard errors: MLE

```
#Adding offset (2)
pois2b<- glm.nb(art ~ fem + mar,
        offset(log(phdage)),
        data = articlesdata)
jtools::summ(pois2b, confint=T, exp=T, mode
```

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 1.89 | 1.80 | 2.00 | 23.90 |
| **femWomen** | 0.79 | 0.73 | 0.85 | -6.05 |
| **marSingle** | 0.97 | 0.89 | 1.05 | -0.78 |

Standard errors: MLE

```
mean(articlesdata$art[articlesdata$fem=="Men" & articlesdata$mar=="Married"])
```

```
## [1] 1.863517
```

# Which Poisson model is preferred - analytical methods

For regular maximum likelihood models:

- **Akaike information criterion (AIC)**, which can be extracted from the associated `glm` model, is often used to compare models with smaller values representing a better fit

*AIC* = *deviance* + *2k*, where k = number of coefficients and deviance = −2 times the log score (`logLik(model)`)

```
cbind(Model= c("Risk", "Rate-Poisson", "Rate-Neg.Binomial (1)",  "Rate-Neg.Binomial (2)"),
    AIC= c(round(pois1$aic, 2), round(pois2$aic, 2), round(pois2a$aic, 2), round(pois2b$aic,
```

```
##      Model                     AIC
## [1,] "Risk"                    "3467.87"
## [2,] "Rate-Poisson"            "3489.55"
## [3,] "Rate-Neg.Binomial (1)" "3226.9"
## [4,] "Rate-Neg.Binomial (2)" "11352.82"
```

Can us Likelihood Ratio test as well, but consider how parsimonious the model can be

# Rate estimation without models

Let's consider sex assigned at birth (SAB) as contrast and the PhD student's age as the "time-exposure" and compare the rate of articles by SAB.

Can use `poisson.test` function from the `stats` package to perform exact Poisson tests (event rate NHT) and comparison of Poisson rates (rate ratios NHT).

```
#NHT for the rate ratio
poisson.test(c(619,930), c(14567,17002))
```

```
##
##     Comparison of Poisson rates
##
## data:  c(619, 930) time base: c(14567, 17002)
## count1 = 619, expected count1 = 714.76, p-valu
## alternative hypothesis: true rate ratio is not
## 95 percent confidence interval:
##  0.7006155 0.8609170
## sample estimates:
## rate ratio
##  0.7768508
```

```
#Rate among Women: 4.2%--> 0.0424
r.fem<-sum(articlesdata$art[articlesdata$fe
  sum(articlesdata$phdage[articlesdata$fem=
#Rate among Men: 5.5%--> 0.0546
r.male<-sum(articlesdata$art[articlesdata$f
  sum(articlesdata$phdage[articlesdata$fem=
# Rate Ratio= 0.776
#
r.fem/r.male
```

```
## [1] 0.7770859
```

# Rate estimation without models

One could also set thresholds and ask whether the rates are different, larger or smaller from...

```
poisson.test(c(619,930), c(14567,17002), r=0.6, alternative = "less")
```

```
##
##     Comparison of Poisson rates
##
## data:  c(619, 930) time base: c(14567, 17002)
## count1 = 619, expected count1 = 546.61, p-value = 1
## alternative hypothesis: true rate ratio is less than 0.6
## 95 percent confidence interval:
##  0.0000000 0.8470421
## sample estimates:
## rate ratio
##  0.7768508
```

If you only have one variable or want to test/reproduce results from publications, could be useful. Otherwise, would have to use the data and fit a model

...The Risk Estimation part!

# Poisson approximation to Binomial

- Cohort studies where all patients have equal follow-up times.
- Poisson regression can be used similarly as logistic regression, with a time-at-risk value specified as 1 for each subject.
- If the model adequately fits the data, this provides a correct estimate of the Adjusted Risk Ratio

Chen et al.: Comparison of robustness to outliers between robust poisson models and log-binomial models when estimating relative risks for common binary outcomes: a simulation study. BMC Medical Research Methodology 2014 14:82.

# Poisson Model

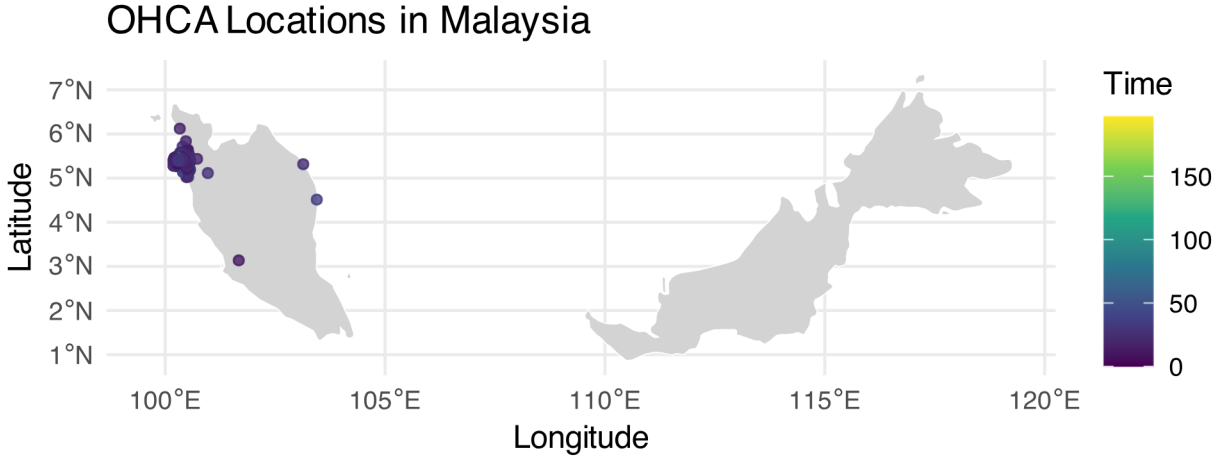
Recall this table?

| Model | Frequentist Estimates | Bayesian Estimates |
|---|---|---|
| Logit Model (Predicted probs.) [2] | RD= 0.073; RR= 1.599 | RD= 0.073; RR= 1.598 |
| AME (prediction `StdGLM`) [2] | RD= 0.073; RR= 1.549 | - |
| AME (`marginaleffects`) [2] | RD= 0.073 ; RR= 1.549 | - |
| AME (marginal standardization) [2] | RD= 0.044 ; RR= 1.333 | RD= 0.073 ; RR= 1.548 |
| GLM: Log-Binomial [2] | RD= ?; RR= 1.493 or RRa= 1.493 | RD= 0.073; RR= 1.488 |

We can add an extra row with estimations using Poisson Regression

# Building a regression model - Out of hospital cardiac Arrest in Malaysia

Example from: Emily Leong, K. E., Benjamin Leung, K. H., Barry, T., & Leong, Y. C. (2026). Establishment of an out-of-hospital cardiac arrest registry in Penang, Malaysia. Resuscitation plus, 31, 101435.



## Out of hospital cardiac Arrest in Malaysia

**Background:** In 2023 an out-of-hospital cardiac arrest (OHCA) registry was established in Penang, Malaysia. Objective: To facilitate systematic, sustainable collection and analysis of OHCA cases across the state. Authors describe the data collection processes and presents a descriptive analysis of first-year data, from April 2023 to March 2024 (n=5087).

In total, 16.3% had a resuscitation attempt by EMS. Most individuals were of either Chinese (52.2%) or Malay (33.0%) ethnicity; most OHCA occurred at home (92.8%). The median ambulance response time was 21 min [IQR 16-29]. Among EMS-attempted resuscitation cases, 47.2% were witnessed by laypersons, bystander CPR was provided in 43.5%, and an AED was applied in 7.1% and the 30-day survival was 1.3%.

OHCA **survival rates** in Penang were low despite **bystander CPR** being performed in nearly half of EMS-attempted resuscitation cases. Prolonged **EMS response times** and low rates of **AED deployment** represent targets for intervention.

## Key Variables:

- **Outcome:** $y_i$ = 1 if individual $i$ was **treated by EMS** or = 0 if did not receive EMS treatment. Potential independent (predictor) variables are:
  • RTmins1: Ambulance Response Time (in minutes)
  • Age: Age of OHCA's case
  • witnessedB: Witnessed status of the OHCA event
  • Ethnicity, AED use, ROSC, Survival status, etc.

# Read in the data

```r
# Data on OHCA provided earliest in the course
file_common <- here::here("_common.R")  # Specific formatting and functions
source(file_common) # Run common code
str(ohca, list.len = 10, give.attr=F)
```

```
## spc_tbl_ [5,058 × 62] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Number                                          : num [1:5058] 1 2 3 4 5 6 7 8 9 10 ...
##  $ MyPHC Number                                    : chr [1:5058] "myOHCA-07-100014" "myOHCA-07-100
##  $ Ethnicity                                       : chr [1:5058] "M" "C" "C" "C" ...
##  $ Incident Date                                   : POSIXct[1:5058], format: "2023-04-01" "2023-04
##  $ Hour of day                                     : num [1:5058] 13 14 20 8 6 21 22 23 16 18 ...
##  $ Hour category                                   : chr [1:5058] "Daytime" "Daytime" "Evening" "Da
##  $ Day of week                                     : num [1:5058] 6 6 6 6 6 6 6 6 6 7 ...
##  $ Week category                                   : chr [1:5058] "Weekend" "Weekend" "Weekend" "We
##  $ EMS Treated                                     : chr [1:5058] "Yes" "No" "No" "No" ...
##  $ EMS witnessed                                   : chr [1:5058] "Yes" "No" "No" "No" ...
##   [list output truncated]
```

# Data overview

```
#summary(ohca) # details can found with ?oh
ohca %>% dplyr::select(Age, Sex, EMStreat,R
```

| Characteristic | N = 5,058[1] |
|---|---|
| Age | 74 (61, 83) |
| Unknown | 28 |
| Sex | |
| Female | 2,288 (45%) |
| Male | 2,765 (55%) |
| Unknown | 5 (<0.1%) |
| EMStreat | 809 (16%) |
| RTmins1 | 21 (16, 29) |
| witnessB | 1,856 (37%) |
| dead | 5,014 (99%) |

[1] Median (Q1, Q3); n (%)

How many individuals with OHCA were treated by EMS?

```
prop.table(table(ohca$EMStreat))
```

```
#>
#>    0    1
#> 0.84 0.16
```

```
# Mean and Variance
mean(ohca$EMStreat)
```

```
#> [1] 0.16
```

```
var(ohca$EMStreat)
```

```
#> [1] 0.134
```

# Poisson Model for Risk Ratios

```
pois4 <- glm(EMStreat ~ witnessB + Age, data=ohca,
             family = poisson(link="log"))
#tbl_regression(pois4,  exponentiate = T)
jtools::summ(pois4,  exp = T, confint=T,  model.fit = F, model.info = F, pvals = F)
```

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 0.75 | 0.60 | 0.94 | -2.49 |
| **witnessB** | 1.55 | 1.35 | 1.78 | 6.20 |
| **Age** | 0.97 | 0.97 | 0.98 | -15.63 |
| Standard errors: MLE | | | | |

Note: there is no offset here!

# Poisson Model for Risk Difference

```
pois5 <- glm(EMStreat ~ witnessB + Age, data=ohca,
             family = poisson(link="identity"))
#tbl_regression(pois4,  exponentiate = T)
jtools::summ(pois5,  exp = T, confint=T,  model.fit = F, model.info = F, pvals = F)
```

Convergency issues... as well!

```
Error: no valid set of coefficients has been found: please supply starting values

 5. stop("no valid set of coefficients has been found: please supply starting values",
    call. = FALSE)
 4. glm.fit(x = structure(c(1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
    1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
    1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
    1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
 3. eval(call(if (is.function(method)) "method" else method, x = X,
    y = Y, weights = weights, start = start, etastart = etastart,
    mustart = mustart, offset = offset, family = family, control = control,
    intercept = attr(mt, "intercept") > 0L, singular.ok = singular.ok))
 2. eval(call(if (is.function(method)) "method" else method, x = X,
    y = Y, weights = weights, start = start, etastart = etastart,
    mustart = mustart, offset = offset, family = family, control = control,
    intercept = attr(mt, "intercept") > 0L, singular.ok = singular.ok))
 1. glm(EMStreat ~ witnessB + Age, data = ohca1, family = quasipoisson(link = "identity"))
```

Luckly you learnt how to handle this :)

# Poisson Model

- Poisson models to get the RISK ratio this way produces biased confidence intervals
- The model assumes a Poisson distribution when in fact the outcomes are binary.
- To overcome issues with the variance, we could use a variance estimator that is robust to the misspecification of the model.

### Why and When do we use Poisson?

- Used for rare outcomes
- When the follow-up has different lengths

# Why and When do we use Poisson?

- For common outcomes, Poisson regression is likely to compute conservative CIs, suggesting less precision than is true.

- Poisson regression produces wider CIs (compared with a log-binomial model and stratified analysis) because Poisson errors are overestimates of binomial errors when the outcome is common

- Poisson errors approximately equal binomial errors when the outcome is rare

**... A solution can be??**

# Robust Poisson Models

The Poisson regression uses a logarithm as the natural link function under the GLM framework.

- The robust Poisson regression model uses the **classical sandwich estimator** under the generalized estimation equation (GEE) framework to correct the inflated variance (also known as over-dispersion) in the standard Poisson regression.

- The technique is known as modified Poisson regression or pseudo-likelihood estimation.

- This correction can be achieved by using the REPEATED statement in SAS Proc GENMOD or the ROBUST option in STATA's Poisson procedure, or `quasipoisson` family and/or `robust` option in `R`.

# Poisson Model for Risk Ratios

```
pois4 <- glm(EMStreat ~ witnessB + Age, dat
             family = poisson(link="log"))
jtools::summ(pois4, exp = T, confint=T,
             robust = "HC0",
             model.fit = F, model.info = F,
```

| | exp(Est.) | 2.5% | 97.5% | z val. |
|---|---|---|---|---|
| **(Intercept)** | 0.75 | 0.63 | 0.90 | -3.15 |
| **witnessB** | 1.55 | 1.37 | 1.75 | 6.99 |
| **Age** | 0.97 | 0.97 | 0.98 | -19.59 |

Standard errors: Robust, type = HC0

```
pois4a <- glm(EMStreat ~ witnessB + Age, da
          family =
          quasipoisson(link = "log"))
jtools::summ(pois4a,  exp = T, confint=T, m
             model.fit = F, pvals = F)
```

| | exp(Est.) | 2.5% | 97.5% | t val. |
|---|---|---|---|---|
| **(Intercept)** | 0.75 | 0.62 | 0.92 | -2.76 |
| **witnessB** | 1.55 | 1.37 | 1.75 | 6.87 |
| **Age** | 0.97 | 0.97 | 0.98 | -17.33 |

Standard errors: MLE

**Check the assumptions!**

# Robust Poisson models vs. Log-Binomial

In presence of linear confounders, the two models yield comparable relative biases

- With the non-linear confounders, the robust Poisson model outperform the log-binomial model.
    - Larger differences with rare outcomes
- The robust Poisson models are more robust to outliers compared to the log-binomial models when estimating relative risks or risk ratios for common binary outcomes.

# Log-Binomial vs Robust Poisson?

**Recall:** Software utilizes iterative weighted least squares (IWLS*) approach or variations of IWLS to find MLEs for generalized linear models.

- For log-binomial models, the weights used by the IWLS approach contain the term $1/(1-p)$, where $p = exp(XT\beta)$ with a range from 0 to 1.
- The MLE of a log-binomial model is likely to be **too sensitive to outliers** because a very large $p$ has a large influence on the weights.
- MLE and pseudolikelihood estimators are deteriorated in presence of outliers.

The level of deterioration differed when the relationships between the confounder and the outcome was not in a simple form.

# Log-Binomial vs Robust Poisson?

- The log-binomial method is generally preferred due to the fact that the MLEs estimated by the log-binomial models are more efficient compared to the pseudo-likelihood estimators used by the robust Poisson models.

- Spiegelman and Hertzmark recommend using the log-binomial models over the robust Poisson models when convergence is not an issue.

- It appears that the gain in efficiency is beneficial to log-binomial models only for samples of small sizes.

Due to the concern of lack of efficiency for the robust Poisson models for small samples, log-binomial may still be the choice when the sample size is small.

# The Bayesian way?

## Risks Ratio

```
pois6<-stan_glm(EMStreat ~ witnessB + Age,
        family = poisson(link="log"), refre
tbl_regression(pois6, intercept=T, exponent
```

| Characteristic | IRR | 95% CI |
|---|---|---|
| (Intercept) | 0.75 | 0.61, 0.93 |
| witnessB | 1.55 | 1.35, 1.78 |
| Age | 0.97 | 0.97, 0.98 |

Abbreviations: CI = Credible Interval, IRR = Incidence Rate Ratio

## Risk Difference

```
pois7<-stan_glm(EMStreat ~ witnessB + Age,
        family = poisson(link="identity"),
tbl_regression(pois7, intercept=T)
```

| Characteristic | Beta | 95% CI |
|---|---|---|
| (Intercept) | 0.38 | 0.34, 0.42 |
| witnessB | 0.04 | 0.02, 0.07 |
| Age | 0.00 | 0.00, 0.00 |

Abbreviation: CI = Credible Interval

# Poisson Models for Risks Ratios and Risks Differences

| Model | Frequentist Estimate | Bayesian Estimates |
|---|---|---|
| Logit Model (Predicted probs.) | RD= 0.073; RR= 1.599 | RD= 0.073; RR= 1.598 |
| AME (prediction StdGLM) | RD= 0.073; RR= 1.549 | - |
| AME (marginaleffects) | RD= 0.073 ; RR= 1.549 | - |
| AME (marginal standardization) | RD= 0.044 ; RR= 1.333 | RD= 0.073 ; RR= 1.548 |
| GLM: Log-Binomial | RD= ?; RR= 1.493 | RD= 0.073; RR= 1.488 |
| GLM: Poisson Regression | RD= ? RR=1.55 | RD=0.04; RD=1.55 |

# Goodness of Fit for Bayesian Models

## Poisson

```
pois6<-stan_glm(EMStreat ~ witnessB + Age,
        family = poisson(link="log"), refre
tbl_regression(pois6, intercept=T, exponent
```

| Characteristic | IRR | 95% CI |
|---|---|---|
| (Intercept) | 0.75 | 0.61, 0.93 |
| witnessB | 1.55 | 1.35, 1.76 |
| Age | 0.97 | 0.97, 0.98 |

Abbreviations: CI = Credible Interval, IRR = Incidence Rate Ratio

## Negative Binomial

```
pois8<-stan_glm(EMStreat ~ witnessB + Age,
        family = neg_binomial_2(link = "log
tbl_regression(pois8, intercept=T, exponent
```

| Characteristic | exp(Beta) | 95% CI |
|---|---|---|
| (Intercept) | 0.76 | 0.60, 0.95 |
| witnessB | 1.55 | 1.34, 1.78 |
| Age | 0.97 | 0.97, 0.98 |

Abbreviation: CI = Credible Interval

# Goodness of Fit for Bayesian Models

### Poisson vs. Negative Binomial

- Can use `loo_compare()` in `rstanarm` package to compare models by `loo`, an approximate leave-one-out cross-validation shortcut Gelman A, ROS page 174

```
loo_compare(loo(pois6),loo(pois8))
```

```
#>  model elpd_diff se_diff p_worse diag_diff diag_elpd
#>  pois6       0.0     0.0      NA
#>  pois8      -9.0     0.6    1.00
```

- Other option is the **Widely Applicable Information Criterion (AIC)** using `waic(modelname)`

```
cbind(Model=c("Poisson", "Neg. Binomial"), WAIC=c(round(waic(pois6)$waic,2), round(waic(pois
```

```
#>      Model           WAIC
#> [1,] "Poisson"       "4324.52"
#> [2,] "Neg. Binomial" "4342.59"
```

# Poisson for Risk Ratios and Differences... Is it a good option?

**It depends!**

- Assumptions
- Common vs Rare Outcomes
- Overdispersion
- Standard Errors
- Convergence

QUESTIONS?

COMMENTS?

RECOMMENDATIONS?

# Mean and Variance relationship by sample

| Distribution | Variance to mean [a] | Sample situation |
|---|---|---|
| Normal | Constant $\sigma^2$ | Linear regression |
| Binomial | $\sigma^2 = n\mu(1-\mu)$ | Successes out of $n$ trials |
| OD [b] Binomial | $\sigma^2 \propto n\mu(1-\mu)$ | Clustered success data |
| Poisson | $\sigma^2 = \mu$ | Count data, variance equals mean |
| OD Poisson | $\sigma^2 \propto \mu$ | Count data, variance proportional to mean |
| Negative binomial | $\sigma^2 = \mu + \mu^2/k$ | Count data, variance quadratic in the mean |
| Gamma | $\sigma \propto \mu$ | Continuous data, standard deviation proportional to mean |

[a] Mean is denoted by $\mu$ and the variance by $\sigma^2$. [b] Over-dispersed.

Vittinghoff E., Glidden D.V., Shiboski S.C., McCulloch C.E. (2012) Generalized Linear Models. In: Regression Methods in Biostatistics. Statistics for Biology and Health. Springer, Boston, MA

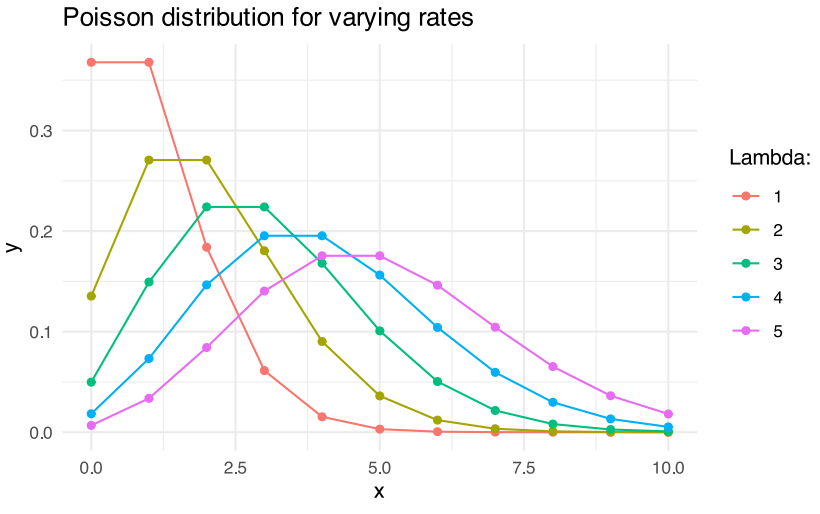
# Plotting Poisson distributions

`ggplot` code

```
# Build Poisson distributions
library(tidyverse)
p_dat <- map_df(1:5, ~ tibble(
  l = paste(.),
  x = 0:10,
  y = dpois(0:10, .)
))

# Use ggplot2 to plot

ggplot(p_dat, aes(x, y,
      color = factor(l, levels = 1:5))) +
  geom_line() +
  geom_point(data = p_dat, aes(x, y,
  color = factor(l, levels = 1:5))) +
  labs(color = "Lambda:") +
  ggtitle("Poisson distribution for varying
  theme_minimal()
```




- As λ increases Poisson distribution approximates to a normal distribution.
- When *n* is large and *p* is very small, Poisson distribution approximates the binomial distribution.

# Maximum likelihood estimate of lambda

$$MLE = \ell(\lambda; y_1, \ldots, y_n) = \sum_{i=1}^{n} \log f(y_i; \lambda) = \sum_{i=1}^{n} \log \frac{\lambda^{y_1} exp^{-\lambda}}{y_1!}$$

Take first derivative (notation $\prime = \frac{d(.)}{d\lambda}$)

$\ell'(\lambda; y_1, \ldots, y_n) = \sum_{i=1}^{n} \log \frac{\lambda^{y_1} exp^{-\lambda}}{y_1!}'$

$= \sum_{i=1}^{n} \{-\lambda + y_i \cdot \log(\lambda) - \log(y_i!)\}'$

$= \sum_{i=1}^{n} \left\{-1 + y_i \cdot \frac{1}{\lambda}\right\}$

$= -n + \frac{1}{\lambda} \sum_{i=1}^{n} y_i$

# Maximum likelihood estimate of lambda

The first-order condition for maximizing the log-likelihood sets its derivative to zero

$$\ell'(\lambda; y_1, \ldots, y_n) = 0$$

$$-n + \frac{1}{\lambda} \sum_{i=1}^{n} y_i = 0$$

$$n \cdot \lambda = \sum_{i=1}^{n} y_i$$

$$\lambda = \frac{1}{n} \sum_{i=1}^{n} y_i = \bar{y}$$

Thus, the maximum likelihood estimator is simply the empirical mean $\hat{\lambda} = \bar{y}$